QBUS6840 Predictive Analytics

(S2 2024)

________540404120___________

Executive Summary

This report presents a Consumer Price Index (CPI) forecasting model to support national economic and monetary policy planning under volatile conditions. Using quarterly historical CPI data from 1990 to 2022, the project examines and evaluates the methodology and forecasting accuracy of several forecasting models, including the trend-adjusted exponential smoothing model (TCES), Auto-Regressive Integrated Moving Average (ARIMA), and Long Short-Term Memory (LSTM), a special case for Recurrent Neural Networks model (RNN). After comparison, the ARIMA (2,1,0) model with intercept gives the relatively most accurate forecast, with a mean squared error (MSE) of 6.47 for the validation set, which indicates the model is suitable for short-term forecasting of CPI.

The ARIMA (2,1,0) with intercept forecast results for the next six quarters is consistent with current economic expectations, supporting its use for forecasting in high cash rate environments. Future improvements may include integrating external indicators, such as cash rates and commodity prices, to improve the CPI forecasting model's responsiveness to economic changes.

Background

The Consumer Price Index (CPI) measures the total change in consumer prices over time based on a representative basket of goods and services (Fernando, 2024). The U.S. Bureau of Labor Statistics (BLS) calculates the CPI as a price-weighted average of a representative set of goods and services that reflect overall consumer spending in the U.S. (U.S. Bureau of Labor Statistics, 2024). Financial market participants widely use the CPI as a measure of inflation. Central banks and governments use the CPI to adjust monetary policy and the amount of social security spending. In addition, because the CPI measures changes in consumer purchasing power, it is often a key factor in wage negotiations (Fernando, 2024).

The task involves creating a reasonably accurate forecasting model using historical quarterly data from March 1990 to December 2022. The model aims to forecast CPI values from March 2023 to June 2024. The economy after 2021 is unstable. The Omicron variant's rapid spread suggested that the pandemic would continue to disrupt economic activities in the short term. The combination of new COVID-19 outbreaks, ongoing supply-chain issues, inflationary pressures, and global financial vulnerabilities raised the risk of a severe economic downturn (World Bank, 2022). In this volatile economic environment, accurate forecasting of the CPI is critical in guiding the central bank in regulating policy through interest rate adjustments and stabilizing the price level and the economy.

We evaluate the accuracy of our model's predictions by using Mean Squared Error (MSE). It quantifies the average squared difference between the actual values and the predicted values, providing insight into how well our model performs.

Mathematically, MSE is defined as:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(CPI_i - \widehat{CPI}_i)^2$$

In addition to presenting the model's accuracy through quantitative research, the report needs to discuss the appropriateness of forecasting CPI values through qualitative assessment. This requires a thorough assessment of the reasonableness of the expected economic trends predicted by the model, using domain knowledge and comparing historical and forward-looking data on cash rates.

Data understanding and cleaning

Data preprocessing starts with loading the dataset to understand it better and prepare for subsequent modeling. The index columns are then specified as date-time indexes to read the data in a time-series fashion. First, the data type of the quarterly column of the CPI dataset is converted to DateTime, and then the quarterly column is set as an index column.

After data analysis, the CPI dataset contains 132 data, with a mean value of 83.6341 and maximum and minimum values of 125.6000 and 50.3000, respectively. This indicates that this dataset is relatively uniformly distributed, with no extremely large or small values, and does not need to be scaled. In addition, there are no null variables in the dataset, eliminating the need to fill in missing values. Detailed information is shown in Table 1.

Table 1: CPI Descriptive Statistics

	CPI
count	132.0000
mean	83.6341
std	21.5584
min	50.3000
25%	62.0750
50%	85.6500
75%	103.8000
max	125.6000
Number of null variables	0.0000

Exploratory Data Analysis (EDA)

The time-series plot in Figure 1 shows a clear upward trend, demonstrating that the CPI has risen steadily since 1990. This reflects that the price level of goods and services increased over the 22-year period studied. Besides, this upward trend was relatively flat and stable until 2021, after which the CPI accelerated rise. This might be because, in the mid-2020s, the pandemic lockdown gradually came to an end, inflation began to pick up as demand and oil prices rebounded and supply disruptions deepened, with global inflation in July 2022 reaching its highest level since the mid-1990s (Ha et al., 2023).

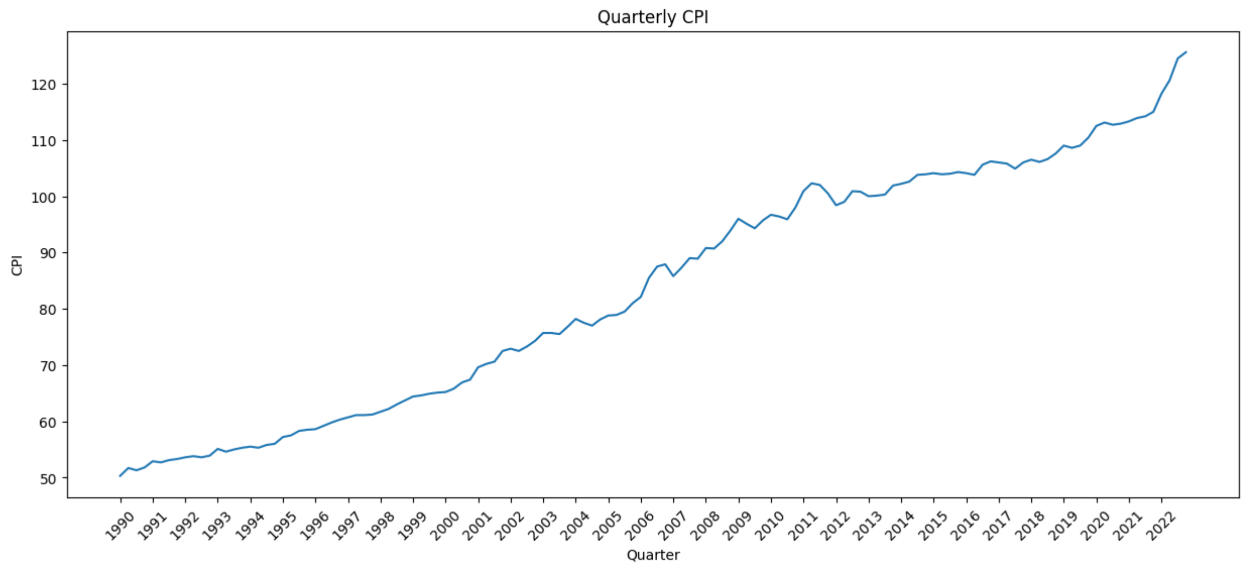

Figure 1: CPI plot

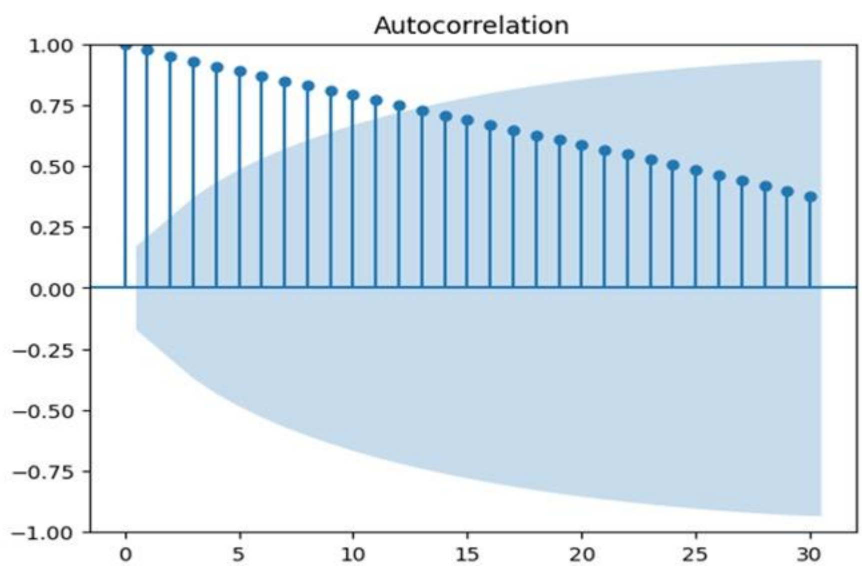

Figure 2: Autocorrelation Function Plot (ACF) from raw data

Next, the stationarity of this dataset is tested. The test begins with a time-series plot and an ACF plot. A time series process is stationary if its mean, variance, and covariance functions remain constant over time. In other words, time series with trends or seasonality, are not stationary — the trend and seasonality will affect the value of the time series at different times (Hyndman & Athanasopoulos, 2018). Since there is a steadily increasing trend over time (Figure 1), our CPI data is non-stationary. Besides, if the autocorrelation function (ACF) plot of a nonseasonal time series dies down extremely slowly or not at all, the time series should be considered nonstationary (Hyndman & Athanasopoulos, 2018). The ACF plot (Figure 2) shows strong autocorrelation that dies down slowly, also implying a non-stationary time series.

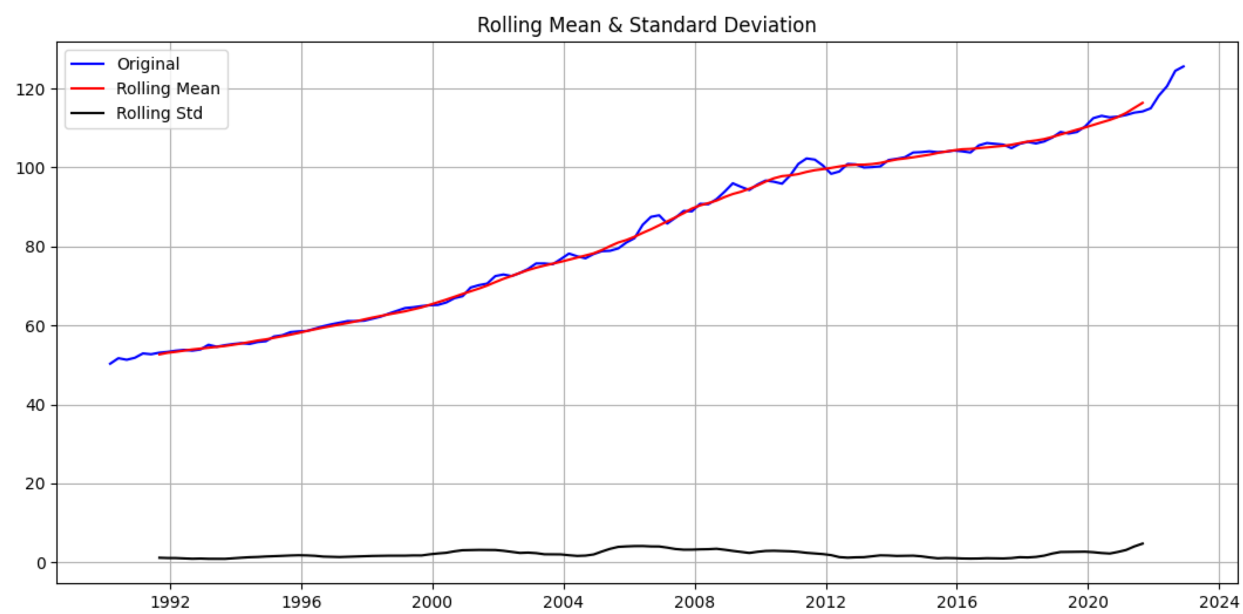

Figure 3: Rolling Mean & Standard Deviation for ADF

The augmented Dickey-Fuller test (ADF) is applied to further verify the stationarity of the CPI time series through hypothesis testing. In Figure 3, the rolling mean of CPI increases even though the standard deviation is small, implying that it is non-stationary.

For ADF, the null hypothesis assumes the presence of a unit root; that is, α=1 makes the time series non-stationary, and the alternative hypothesis is usually stationarity or trend-stationarity (Prabhakaran, 2019). The p-value of our test is 0.9937, which is higher than significant levels at 1%, 5%, and 10%. Therefore, the null hypothesis (H_0) that the time series is non-stationary cannot be rejected.

Methodology

Before modeling, the data is divided into training and validation groups. Studies, including Gholamy et al. (2018), suggest that the best modeling results are achieved when 20-30% of the data is allocated for testing and the remaining 70-80% for training. Thus, 80% of the CPI dataset was allocated as training data for modeling and the remaining 20% as validation data to calculate the MSE to test the model prediction accuracy.

Since this time series data lacks obvious seasonality and cyclicality and only shows a significant upward trend, a Naive forecast, Trend-Adjusted Exponential Smoothing (TCES), Auto-Regressive Integrated Moving Average (ARIMA) model, and RNN (LSTM) model are used to predict the CPI.

For naïve forecasts, all forecasts are simply set to the value of the last observation $\widehat{CPI}_{t+1|1:t} = CPI_t$ (Hyndman & Athanasopoulos, 2018). This method is unsuitable for CPI data with an upward trend and its predictive accuracy would be very low.

The following part will analyze in detail the TCES, ARIMA, and RNN models, which have the smallest MSE.

Model 1: Trend-Adjusted Exponential Smoothing (TCES)

Trend-adjusted exponential smoothing (TCES) is an extension of simple exponential smoothing that includes a level equation (l_t) to reflect the base level and adds a second equation (b_t) to capture the trend in the data. This model aims to forecast time series data with both level and trend components, providing a more accurate forecast. This is beneficial for our data, which shows a systematic increase over time. The framework's ability to generate reliable forecasts quickly and for a wide range of time series is a major advantage and has important implications for industry applications (Hyndman & Athanasopoulos, 2018).

Defining equation:

The statistic model of TCES is defined below:

$$\hat{y}_{t+h|t} = l_t + hb_t$$

Level equation:

$$l_t = \alpha y_t + (1-\alpha)(l_{t-1}+b_{t-1})$$

Trend equation:

$$b_t = \beta(l_t - l_{t-1}) + (1-\beta)b_{t-1}$$

Holt’s linear model is used to process the forecast for CPI, which is defined below:

$$\widehat{CPI}_{t+h|t} = l_t + hb_t$$

For example, for a one-step-ahead forecast ($h = 1$), the forecast equation should be:

$$\widehat{CPI}_{t+1|1:t} = l_t + b_t = \alpha y_t + (1-\alpha)(l_{t-1}+b_{t-1}) + \beta(l_t - l_{t-1}) + (1-\beta)b_{t-1}$$

Where α ($0 \leq \alpha \leq 1$) is the smoothing parameter for the level, controlling how responsive the model is to changes in the time series level, and β ($0 \leq \beta \leq 1$) is the smoothing parameter for the trend, controlling how quickly the model updates the trend based on new observations. A high value of β better places more weight on recent observations, allowing the trend component to react more quickly to changes in the trend, whereas a low value of β gives less weight to the most recent trends and tends to smooth out the present trend (Profillidis & Botzoris, 2019)

Conduct model:

Running the Exponential Smoothing function from statsmodels.tsa.holtwinters can automatically fit and solve for the optimal hyperparameters (α and β) by minimizing the Sum Squared Error (SSE). Since there is no seasonal pattern for our data, we ignore the ɣ and set 'seasonal' as none. Additionally, the 'trend' is set to be additive since the variation around the trend of the data set doesn't change over time (Figure 1). Before training the model, the initial level (l_0) and the initial trend (b_0) need to be identified. The results given by the model are shown below.

Table 2: summary results of Holt's Linear Model

	Additive
α	1.0
β	1.082644e-13
ɣ	NaN
l_0	4.978269e+01
b_0	5.173078e-01
SSE	8.308885e+01

The value of α is 1, indicating that it only considers the latest data, and each new observation essentially replaces the previous smoothed level value entirely. As a result, the forecasts are very responsive to the latest data, leading to high volatility in cumulative smoothing if the data points fluctuate significantly. The value of β is extremely close to zero (1.082644e-13), suggesting that the model doesn't significantly adjust the trend component over time. Therefore, the trend is almost static and doesn't accumulate much.

The model's choice of hyperparameters, coefficients and performances are examined by plotting forecasted CPI and CPI values from the validation set. In our analysis, the MSE for the validation set was calculated to be 6.4399 (Figure 4).

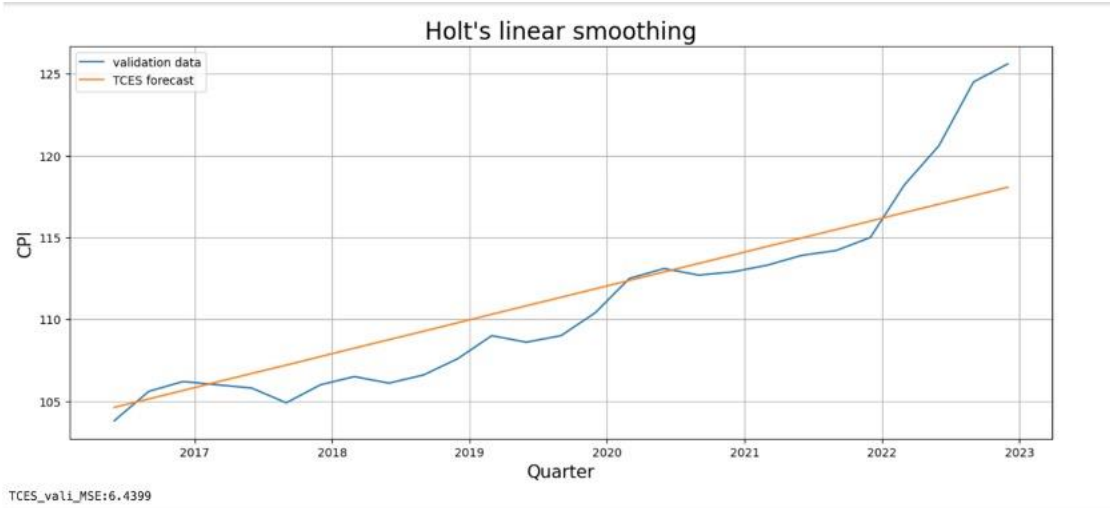

Figure 4: Smoothing and Forecast result

Holt’s Linear Trend Model, while influential in long-term forecasting, does present certain limitations. Notably, the process of determining optimal initial values can complicate and extend the duration of the optimization procedure. Furthermore, when applying smoothing to a dataset over time, it is essential to allocate weights to observations with careful consideration of their temporal position as Holt’s method assumes that the trend is linear and will continue at the same rate into the future. This makes it less practical for some applications where the trend of time series evolves over time. This problem can be seen in our forecasted CPI results, where the forecast values struggle to closely track the validation data during sudden changes. Moreover, the model’s parameters (level and trend) need to be updated frequently, especially when the underlying data behavior changes over time (Winters,1960). These limitations are the reason that alternative approaches were explored.

Model 2: Auto-Regressive Integrated Moving Average (ARIMA) model

Applicability of ARIMA model

In many time series, data might not have a fixed mean over time, which means it is non-stationary. In such cases, it exhibits homogeneity, where different parts of the series behave similarly. To model such behavior, differencing (subtracting consecutive observations over a defined lag) can be applied to transform that series into stationery (Box et al., 2016), making it suitable for using an ARMA (Autoregressive Moving Average) model to analyze.

The ARIMA model extends ARMA by adding a difference step, alongside the autoregressive (AR) and moving average (MA) components. It is well-regarded for its ability to capture complicated underlying patterns in the time series such as autocorrelations, rather than trends and seasonality (Hyndman & Athanasopoulos 2018). The ARIMA has been used to forecast many economic indicators such as Gross Domestic Product (GDP) growth rates, inflation rates, unemployment rates, etc. (Liu, 2024). Studies, such as Nyoni’s (2019) research on CPI forecasting in Germany, have shown that ARIMA models are effective methods for forecasting such economic indicators.

However, it is crucial to note that the model has some limitations. It relies heavily on past information. In other words, it is less flexible in adapting to irregular economic events such as recessions or market crashes. Additionally, the process of selecting hyperparameters can be subjective, potentially affecting the analysis (Liu, 2024).

Conduct a model:

To conduct ARIMA model, the values of hyperparameters p, d, and q need to be determined:

There are 2 ways to select those hyperparameters:

Use the Autocorrelation Function (ACF) plot and The Partial Autocorrelation Function (PACF) plot.

Differencing (d): EDA results indicate the data is non-stationary. When first-order differencing (d = 1) is applied, the ACF plot decays quickly, indicating that d = 1 is appropriate.

Autoregressive Order or **AR(p)**: The PACF plot (Figure 6) reveals a cut-off after lag 2. Thus, setting p = 2 is a suitable option.

Moving Average Order or **MA(q)**: From the ACF plot after differencing (Figure 5), there is a cut-off after lag 2. Therefore, q = 2 is chosen.

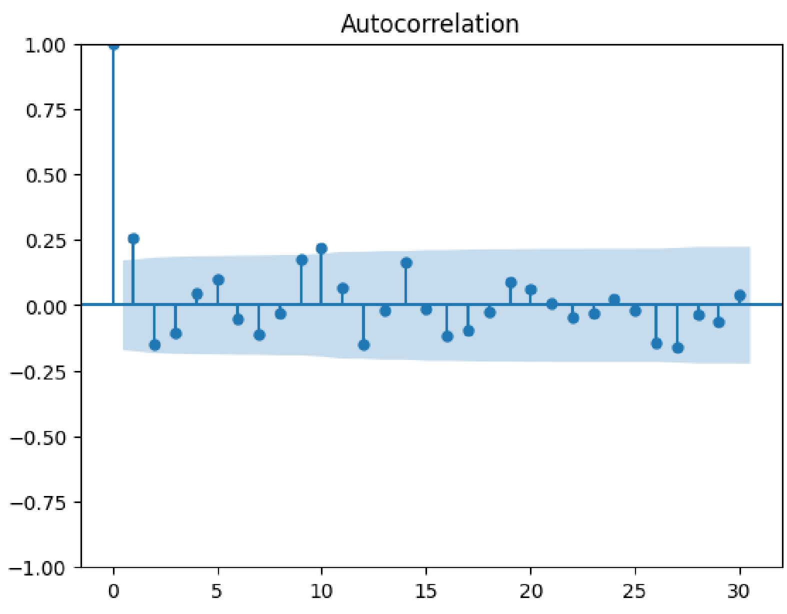

Figure 5: Autocorrelation Function Plot (ACF) after first-order differencing

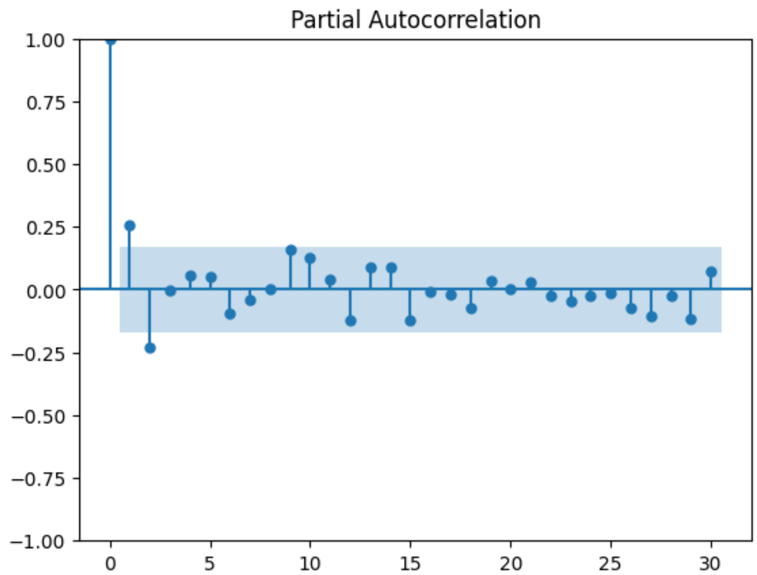

Figure 6: Partial Autocorrelation Function Plot (PACF) after first-order differencing

Use the Auto ARIMA algorithm, which will seek for best value of p,d,q

Maximum p and q are set as 5, the same as the default value from the algorithm. (Sktime developers, 2024). The Auto ARIMA came up with 10 models as Table 3 below. From the Akaike Information Criterion (AIC) value, ARIMA(2,1,0) with intercept has the lowest. According to Hyndman & Athanasopoulos (2018), a lower AIC value indicates a better model.

Table 3: Summary results of what Auto ARIMA algorithm generated

Model	AIC
ARIMA(0,1,0) with intercept	275.7930
ARIMA(1,1,0) with intercept	274.3150
ARIMA(0,1,1) with intercept	270.6240
ARIMA(0,1,0) with intercept	303.8390
ARIMA(1,1,1) with intercept	270.4740

ARIMA(2,1,1) with intercept	267.3820
ARIMA(2,1,0) with intercept	265.4430
ARIMA(3,1,0) with intercept	267.3790
ARIMA(3,1,1) with intercept	269.3790
ARIMA(2,1,0)	289.6420

Table 4: Summary results of ARIMA(2, 1, 0) with intercept

```
                               SARIMAX Results
==============================================================================
Dep. Variable:                      y   No. Observations:                  105
Model:               SARIMAX(2, 1, 0)   Log Likelihood                -128.721
Date:                Mon, 28 Oct 2024   AIC                            265.443
Time:                        20:16:53   BIC                            276.020
Sample:                    03-01-1990   HQIC                           269.728
                         - 03-01-2016
Covariance Type:                  opg
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
intercept      0.5587      0.114      4.915      0.000       0.336       0.782
ar.L1          0.2431      0.077      3.153      0.002       0.092       0.394
ar.L2         -0.3167      0.099     -3.198      0.001      -0.511      -0.123
sigma2         0.6943      0.079      8.795      0.000       0.540       0.849
===================================================================================
Ljung-Box (L1) (Q):                   0.02   Jarque-Bera (JB):                13.65
Prob(Q):                              0.90   Prob(JB):                         0.00
Heteroskedasticity (H):               3.84   Skew:                             0.48
Prob(H) (two-sided):                  0.00   Kurtosis:                         4.49
===================================================================================

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-step).
```

Model Selection using Mean Squared Error (MSE) and Akaike Information Criterion (AIC)

By comparing ARIMA(2,1,2) with ARIMA(2,1,0) with intercept, the MSE and AIC values are as below.

Table 5: Comparison between ARIMA(2,1,2) and ARIMA(2,1,0) with intercept

Model	Train MSE	Validation MSE	AIC
ARIMA(2, 1, 2)	0.7570	6.8361	276.7270
ARIMA(2,1,0) with intercept	0.6964	6.4733	265.4430

ARIMA(2,1,0) with intercept has lower validation MSE and lower AIC values. Thus, it is chosen as the final ARIMA model.

Defining Equation:

Since d = 1, a difference on the original time series Y_t to make it stationary can be conducted. The difference in series (Z_t) can be defined by:

$$Z_t = Y_t - Y_{t-1}$$

Next, AR(2) can be defined by:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t$$

where c is a constant, ϕ_1 and ϕ_2 are the autoregressive parameters, and ϵ_t is white noise.

MA(0). This means the model does not account for any moving average (θ which is moving average parameters will have no values)

By combining the differencing step, AR(2), and MA(0) components, the resulting equation of ARIMA(2, 1, 0) model can be written as:

$$Y_t - Y_{t-1} = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t$$

By integrating the values from summary table, the final equation is:

$$Y_t - Y_{t-1} = 0.5587 + 0.2431 Y_{t-1} - 0.3167 Y_{t-2} + \epsilon_t$$

Trained ARIMA to forecast CPI?

The model was trained by using training data of past CPI to estimate parameters ϕ_1, ϕ_2 θ_1, and θ_2. From table 4, the interpretable values in ARIMA(2,1,0) with intercept are as follows:

Intercept = 0.5587: It suggests that, on average, the CPI changes by approximately 0.5587 units per period, assuming no influence from prior lags.

AR Parameters:

ar.L1, which is the coefficient for the first lag in the autoregressive part (ϕ_1) is 0.2431, indicating that the previous period's change in CPI value has a slight positive influence on the current period's change CPI.

ar.L2, which is the coefficient for the second lag in the autoregressive part (ϕ_2) is -0.3167, indicating that the changes from 2 periods ago have a slight negative effect on the current period's change.

Model 3: Recurrent Neural Networks model (RNN) and Long short-term memory (LSTM)

RNNs are designed to process sequential inputs by maintaining a hidden state that is passed from one-time step to the next (Hochreiter & Schmidhuber, 1997). This hidden state acts like memory, allowing the network to retain information from previous inputs while processing new inputs (Hochreiter & Schmidhuber, 1997).

The structure of RNN includes the input layer (x_t), hidden layer (h_t) and output layer (y_t). At each time step, an input is fed into the network (Goodfellow et. al., 2016). The hidden layer contains a set of neurons, with the output of each neuron at time t dependent on both the current input (x_t) and the previously hidden state (h_t) (Goodfellow et. al., 2016). The network produces an output (y_t) at each time step, which can be used for tasks like prediction or classification (Goodfellow et. al., 2016). RNN's main idea is to let the set of hidden units (h_t) feed on its lagged value (h_{t-1}).

It is important to note that RNN is a general term, not a specific model. It depends on how hidden states are constructed. Then, specific variants of RNN can be defined. The simple recurrent structure is described by the following equations:

$$h_t = \phi(Ux_t + Wh_{t-1} + b)$$

$$\eta_t = \beta_0 + \beta_1 h_t$$

$$y_t = \eta_t + \varepsilon_t$$

Here $\phi(\cdot)$ is an activation function. Using the unfold graph of general RNN and the model, the graphical representation of the Simple RNN model can be plotted as shown in Figure 7 below. The simple RNN model is a sequence of the Simple Recurrent Network (SRN) units whose mathematical equation is described above.

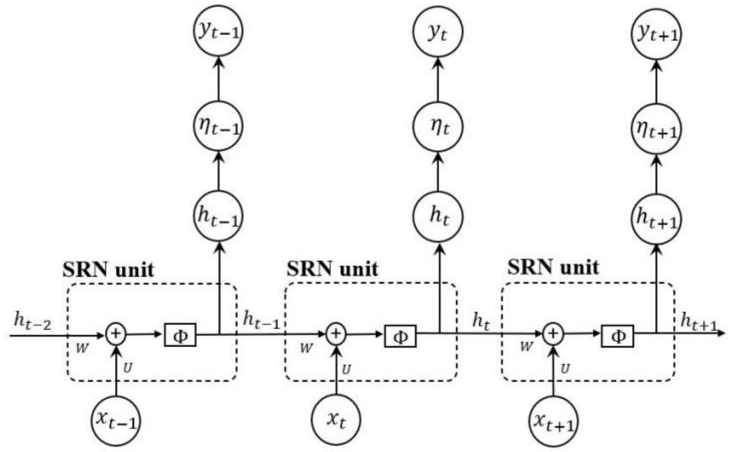

Figure 7: SRN

There are several challenges in RNNs. Due to the repeated multiplication of small or large numbers in the gradient computations, RNNs can suffer from vanishing or exploding gradients, making it difficult to learn long-term dependencies. Standard RNNs also struggle to maintain information over long sequences, which can limit their ability to model long-range dependencies in data.

However, there is an important variant of RNN called Long short-term memory (LSTM). LSTMs are designed to overcome the vanishing gradient problem. They include a memory cell and gating mechanisms, such as input, forget, and output gates, that control the flow of information, enabling the network to retain information over longer sequences (Hochreiter & Schmidhuber, 1997). The main idea of LSTM is to use LSTM units in recurrent neuron networks to capture long-term dependencies and nonlinear effects that may occur in time series. For coding in Python, the Keras library can be used to streamline the changes of neural networks to include LSTM layers.

Before training the model, the appropriate time window, also called the sequence length or look-back window, should be selected for RNNs in time series forecasting. This parameter determines how many past time steps the model considers when making predictions. Choosing the right time window is crucial to capture relevant time dependencies without introducing too much noise or irrelevant information. The ideal time window can often be guided by our knowledge of the problem and the underlying data. Since the quarterly CPI will be forecasted, and there is no seasonal pattern in our time series data, the range of time window has been decided as 1 to 4. Then, a grid search was used to determine the one that has the best performance, which is 1 for our series (Figure 8).

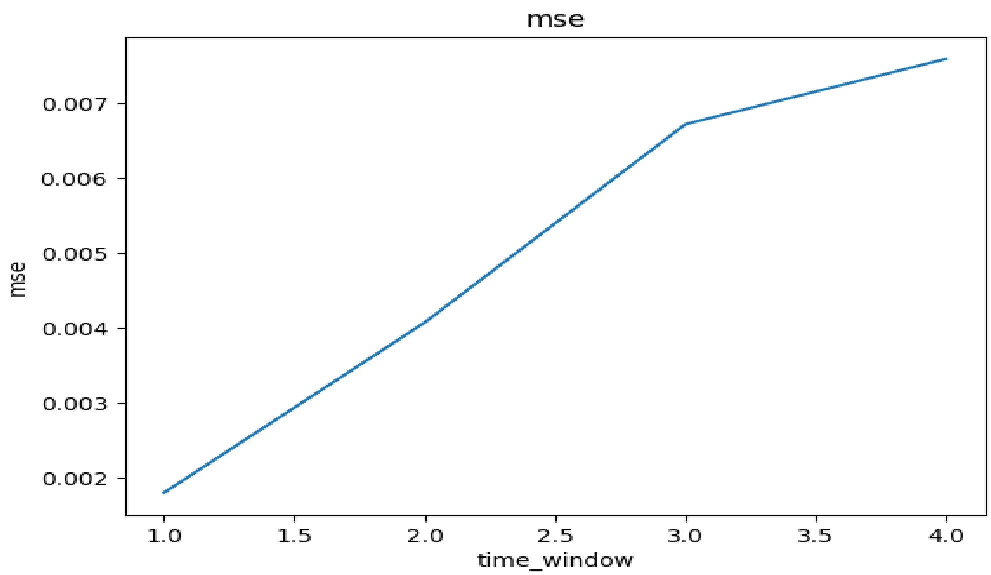

Figure 8: Grid Search for Time Window

Hyperparameter optimization is much more difficult with LSTM since we now have more interacting parameters and a more complex network. Sometimes tuning for the data can take a lot of time. Random search selects random combinations of hyperparameters and evaluates them. It is often more efficient than grid search, as it explores a wider range of the hyperparameter space without testing all combinations (Senapati, 2018). We use Sklearn to process the optimization using random search and determine the following hyperparameters based on the training data.

Units is the dimension of the weight vector inside the LSTM cells (Keras, n.d.). Increasing this number can lead to overfitting and increase the computation costs, but decreasing can lead to poor accuracy. For our dataset, it is 12.

Batch size is the number of samples used per gradient update (Keras, n.d.). Larger batch sizes speed up training but can lead to less precise updates. It is suggested that for time series forecasting, decreasing the batch size we may get less accuracy, but each Epoch completes quicker. The result is 5 for our data.

An **Epoch** is a forward or backward pass of all batches (Keras, n.d.). More epochs give the model more chances to learn but increase the risk of overfitting. We find the optimal one is 500.

The **validation split** is the proportion of data held out for use as a validation set (Keras, n.d.). As it decreases training accuracy will increase since more training data is available, but the guarantee of generalization is much lower. We use the proportion of 10% to make sure we have enough data to train the model.

Shuffle will decide whether to shuffle the training data before each epoch (Keras, n.d.). For time series data, we need to keep it false to ensure the data is in order.

After the optimization, we reconstruct the LSTM model using the optimized hyperparameters. Since our question is a multi-step-ahead forecast, we can use a dynamic forecast using LSTM.

This forecast uses observations up to the last one in the training set. From that point onwards we use a combination of real values and estimated values as input features. When the forecast length exceeds the time window values, which is 1 in our case, we will only use forecast values as inputs.

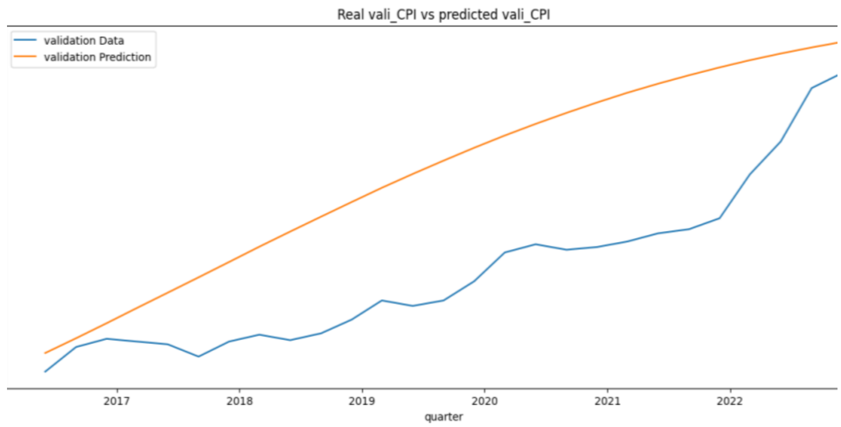

Figure 9: Dynamic Forecast using LSTM

Table 6: MSE of Dynamic Forecast using LSTM

	MSE
Dynamic Forecast using LSTM	61.9284

After we fit our LSTM with the training data, we have the following forecast values and calculate the MSE based on our validation data. The MSE of LSTM for the data with the original scale is 61.9284 (Table 6), which is quite large.

In dynamic forecasting, the model generates forecasts for future time steps based on its own previously predicted values, rather than actual observations (Hochreiter & Schmidhuber, 1997). This recursive process leads to error accumulation (Bengio et. al., 2015), where small errors in earlier predictions are magnified in subsequent predictions, resulting in high MSE.

In addition, although LSTM is a powerful model, it is much more complex and can easily overfit the training data, especially if the dataset is small or contains noise (Goodfellow et. al., 2016), making the model fail to generalize to unseen data and cause a high MSE for the prediction. According to our one-step-ahead forecast of LSTM (Figure 10), the training MSE is 0.5786 (Table 7) while the validation MSE is 1.3930 (Table 7), which is higher than the training MSE, indicating that the model might be overfitting.

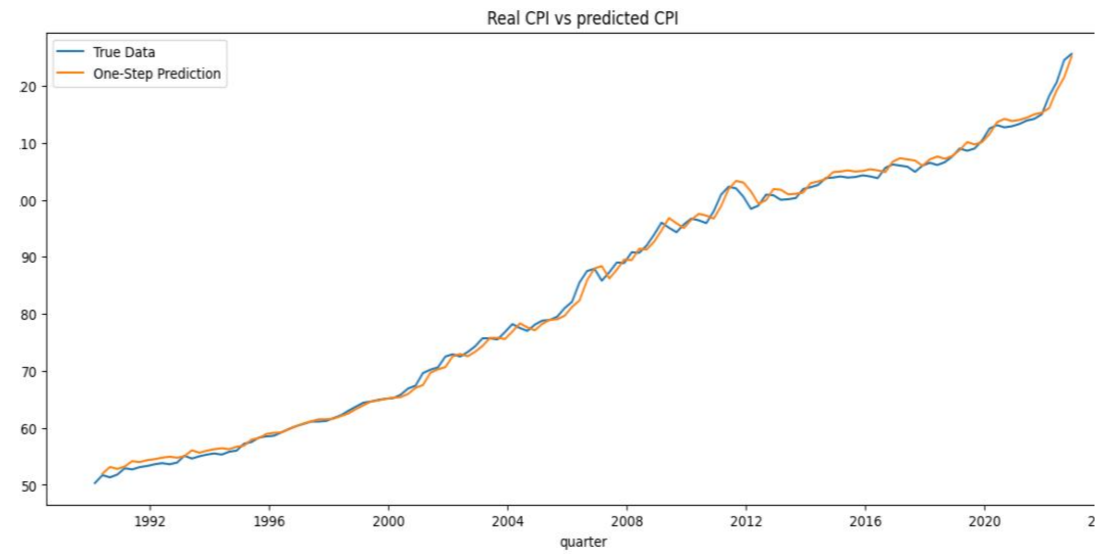

Figure 10: One-step-ahead Forecast using LSTM

Table 7: Training and Validation MSEs of One-step-ahead Forecast using LSTM

	MSE
Training data	0.5786
Validation data	1.3930

Final Model Selection

Among the models tested, including Naive, TCES, LSTM, and ARIMA, the TCES model provided the most accurate predictions, followed closely by the ARIMA model, as shown in Table 8. Although test data was unavailable, a model with a lower mean squared error (MSE) on the validation set is likely to generalize well to unseen data. While the TCES model achieved a slightly lower MSE of 6.4399 compared to the ARIMA model with an MSE of 6.4733, the ARIMA (2,1,0) model with intercept was ultimately selected as the final model due to its interpretability and stability in economic forecasting.

Firstly, ARIMA models are particularly well-suited for time series data like CPI due to their clear statistical structure, which allows for straightforward analysis and interpretability (Box et al., 2016). Unlike the TCES model, which may capture minute nuances in data but lacks an interpretable structure, the ARIMA model's parameters directly relate to observable time-dependent behavior (i.e., trend and seasonality), enabling clearer insight into the data's underlying patterns (Hyndman & Athanasopoulos, 2018). This interpretability is crucial for CPI forecasting, where economic practitioners and policymakers must understand and validate the rationale behind predictions, not just the forecasted values (Hyndman & Athanasopoulos, 2018).

Secondly, ARIMA models tend to exhibit more stability and robustness, particularly when applied to economic data characterized by unexpected shifts and broader fluctuations. In contrast, the TCES model lies in its reliance on a linear combination of level (l_t) and trend (b_t) components to forecast values (y_t), as indicated in the model's underlying equation. This structural design implies that y_t can increase indefinitely over time, a characteristic that often leads to impractical and overly optimistic predictions in extended forecasting scenarios. In cases involving noisy or volatile data, such as CPI, this tendency for values to approach infinity makes the TCES model's forecast less reliable for practical applications (Hyndman & Athanasopoulos, 2018). This robustness factor makes ARIMA more suitable for practical, multi-step-ahead forecasting where stability is paramount, as is the case here for forecasting CPI values up to June 2024.

Thus, the ARIMA model's combination of interpretability, robustness, and reliability in handling time-dependent economic data supports its selection as the final model for forecasting, despite the TCES model's minor MSE advantage.

Table 8: Mean Squared Error of Different Models

Models	**MSE**
Naive	81.7648
TCES	6.4399
ARIMA (2,1,0) with intercept	6.4733
LSTM	61.9284

Results

Quantitative Measure

The ARIMA (2,1,0) model with intercept generated the following CPI predictions in Table 9.These values are expected to be in line with CPI trends, demonstrating moderate growth over the forecasted period, which we can evaluate by comparing with actual values using MSE as they become available.

Table 9: 6 quarterly CPI Predictions

Date	CPI
2023-03-01	125.5017
2023-06-01	125.7235
2023-09-01	126.3505
2023-12-01	127.0273
2024-03-01	127.6180
2024-06-01	128.1684

Qualitative Measure

Beyond quantitative accuracy, it's essential to assess the forecasted CPI values in the broader economic context, specifically by comparing them with cash rate trends. The central bank's cash rate, an influential monetary policy tool, affects inflationary pressures and can help validate or critique our model's predictions.

1. Relationship between CPI and Cash Rate

Generally, a lower cash rate environment is expected to lead to higher inflation as borrowing costs decrease, encouraging spending and investment. However, the effect of cash rate changes on CPI is often delayed, as it takes time for rate adjustments to influence consumer prices fully *(*Reserve Bank of Australia, 2022).

Figure 11 shows CPI and Cash Rate changes from 4 Jan 2011 to 30 Dec 2022. Although the cash rate declined significantly from 2011 to 2022, CPI continued to rise gradually. This suggests that while cash rate reductions can create an inflationary environment, they are not the sole drivers of CPI, as CPI is influenced by a mix of supply-demand factors and global economic conditions (Beckers et. al. , 2023). Around 2021, cash rate reached historically low levels, aligning with the onset of the COVID-19 pandemic. This was a part of emergency measures taken by the central bank to support the economy. However, in 2022, both cash rate and CPI began to increase again, reflecting a shift to control rising inflation pressures post-pandemic. This hints at a potential lagged effect of low cash rates on CPI but also reflects other inflationary factors beyond monetary policy.

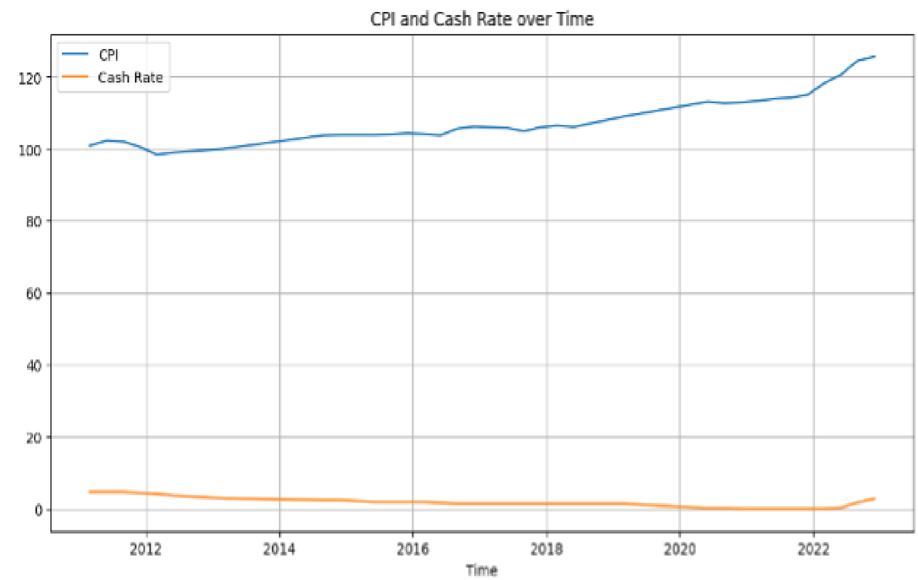

Figure 11: CPI and Cash Rate from 4 Jan 2011 to 30 Dec 2022

2. Cash Rate Outlook and Expected CPI Trends (Jan 2023 – Sep 2024)

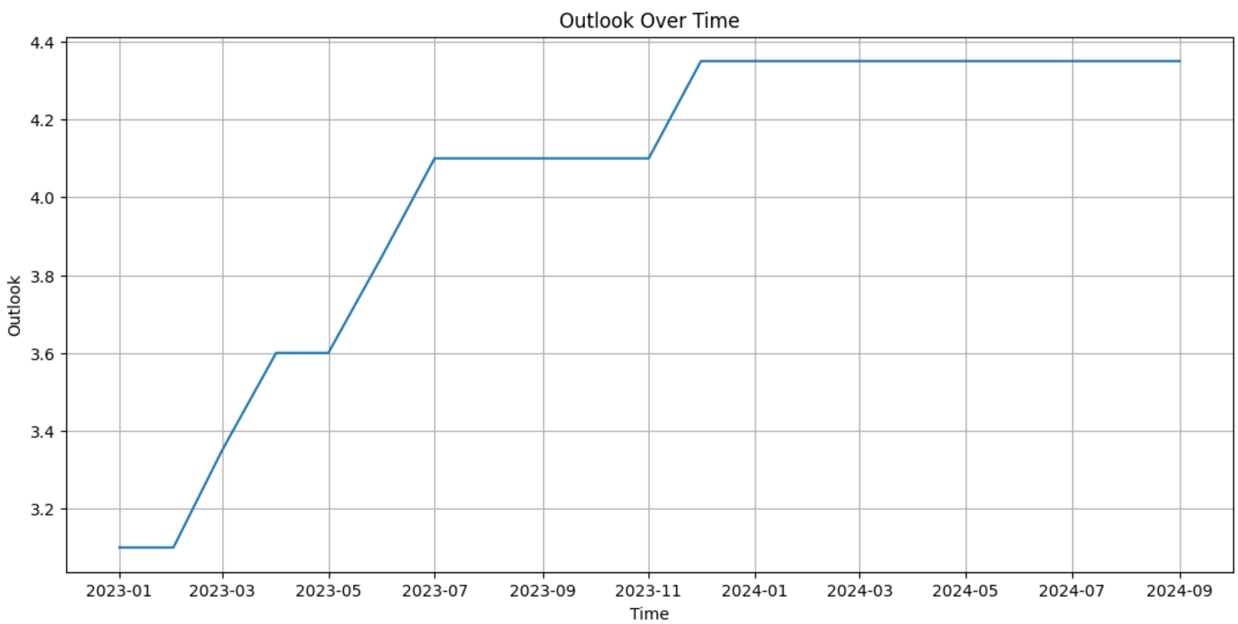

Figure 12: Cash rate outlook from Jan 2023 to Sep 2024

The cash rate outlook data from January 2023 through September 2024 projects a rise to approximately 4.4% in Figure 12, suggesting a restrictive monetary policy environment. This stability suggests an expectation that inflation may moderate, but at a level that justifies maintaining relatively high rates to keep inflation under control. Therefore, the expected trend in CPI growth from 2023 to mid-2024 is likely to be moderate or slightly restrained due to the continued high cash rate environment. The stable cash rate over this period implies that CPI growth should not exhibit drastic changes but rather align with a gradual decline in inflation, aiming for price stability.

3. Consistency of ARIMA Predictions with Economic Expectations

The predicted CPI values from the ARIMA model demonstrate a gradual upward trend, with CPI estimates ranging from 125.5017 in March 2023 to 128.1684 in June 2024. In Figure 13, this steady increase is consistent with expectations given a stable yet high cash rate, which is designed to slow inflation without causing deflationary pressures.

Given the cash rate's projected stability at around 4.0% to 4.4%, the ARIMA model's moderate CPI growth aligns with the economic forecast that inflation will be controlled without significant acceleration. The gradual increase in CPI suggests that the inflationary pressures observed in previous years are expected to ease, as a higher cash rate environment typically dampens the demand side of the economy (Reserve Bank of Australia, 2022). The model's forecasted CPI growth thus aligns with this macroeconomic outlook, projecting that while prices will continue to rise, the pace will be contained by the high-interest rate policy in place.

The consistency between the ARIMA model's predictions and economic expectations supports the validity of the model for CPI forecasting. This consistency further indicates that the ARIMA (2,1,0) model is well-suited for capturing CPI trends under a controlled inflationary environment, where the cash rate is used as a stabilizing instrument. The alignment of the ARIMA-based forecasts with anticipated economic outcomes enhances confidence in the model's utility for CPI projections in a high cash rate context, and this alignment provides a solid basis for using these predictions in policy-related analyses or business decision-making processes that require forward-looking CPI estimates.

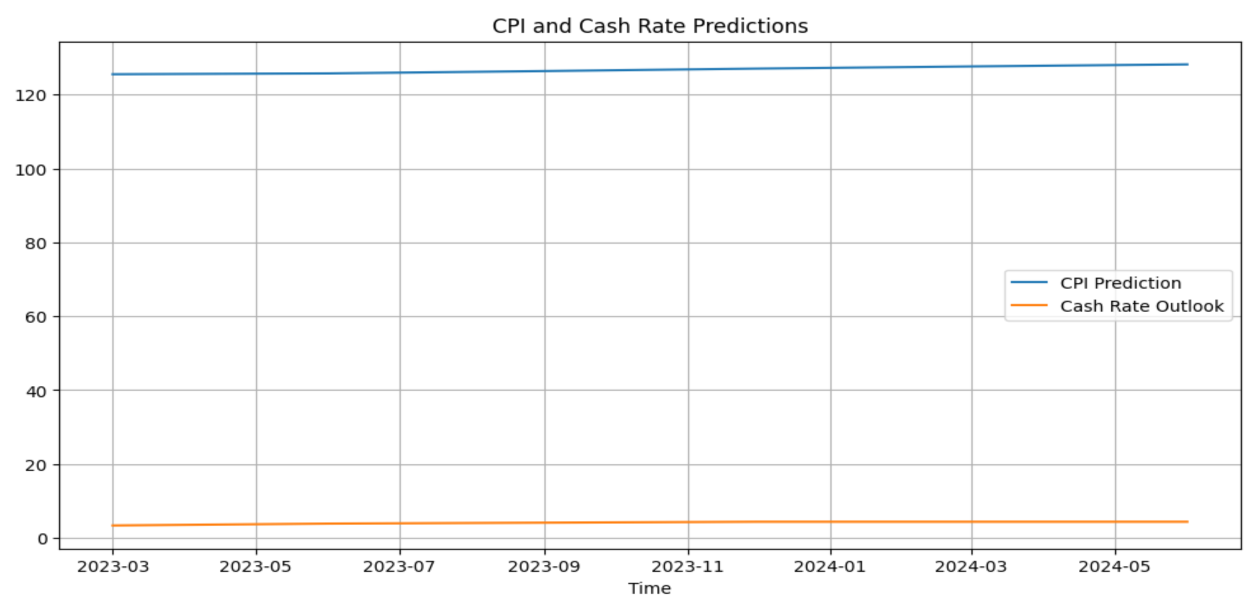

Figure 13: CPI and Cash Rate Predictions

Conclusion

The development of a forecasting model for the Consumer Price Index (CPI) was undertaken to address both quantitative accuracy and economic relevance in the context of historical and forward-looking data. The ARIMA (2,1,0) model with intercept was selected as the final predictive model due to its superior performance in validation, evidenced by its minimal mean squared error (MSE) relative to alternative approaches like Naive, TCES and LSTM. This model's consistent performance suggests a reliable framework for projecting CPI in a volatile economic environment, influenced by factors such as the global pandemic, inflationary pressures, and changing monetary policies. In terms of qualitative analysis, the forecasted CPI trends appear reasonably aligned with recent economic signals, such as the cash rate outlook and recent inflationary trends. It underscores the model's appropriateness in capturing the expected interplay between cash rate adjustments and CPI, given the delayed impact of cash rate changes on consumer prices.

However, the ARIMA model, like any statistical model, has inherent limitations that impact its ability to fully capture CPI dynamics in a highly uncertain environment. ARIMA models, by design, rely heavily on historical trends and assume that future patterns will closely resemble past data behavior. This dependency can limit the model's adaptability to sudden economic shifts or external shocks, such as further pandemic disruptions or unexpected policy changes. Additionally, ARIMA does not explicitly incorporate external economic indicators like cash rates directly within the model, relying instead on CPI historical values alone. This restricts the model's ability to factor in external influences on inflation that could emerge rapidly due to geopolitical factors, global supply chain changes, or fiscal policy shifts.

Future improvements to the CPI forecasting model could involve exploring hybrid models that integrate external indicators, such as the cash rate, unemployment data, or global commodity prices, to enrich the model's responsiveness to economic changes. Moreover, a larger training dataset with daily or monthly data granularity might further enhance model precision and accuracy.

In conclusion, while the ARIMA (2,1,0) model with intercept provides a solid baseline for CPI forecasting, expanding the model's scope to account for real-time economic indicators would enhance its relevance and accuracy, especially in periods of heightened economic volatility. This comprehensive approach to forecasting could yield more reliable guidance for policymakers and businesses in anticipating and responding to future inflationary trends.

References

Beckers, B., Hambur, J., & Williams, T. (2023, June 15). *Estimating the relative contributions of supply and demand drivers to inflation in Australia.* Reserve Bank of Australia. https://www.rba.gov.au/publications/bulletin/2023/jun/estimating-the-relative-contributions-of-supply-and-demand-drivers-to-inflation-in-australia.html#fn0

Bengio, S., Vinyals, O., Jaitly, N., & Shazeer, N. (2015). Scheduled Sampling for Sequence Prediction with Recurrent Neural Networks. NIPS'15: Proceedings of the 28th International Conference on Neural Information Processing Systems (pp. 1171-1179). *MIT Press*. https://doi.org/10.48550/arXiv.1506.03099

Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2016). *Time Series Analysis, Forecasting and Control*. John Wiley & Sons, Inc. http://repo.darmajaya.ac.id/4781/1/Time%20Series%20Analysis_%20Forecasting%20and%20Control%20%28%20PDFDrive%20%29.pdf

Chatfield, C. (2003). *The Analysis of Time Series: An Introduction* (6th ed.). CRC Press.

Fernando, J. (2024, October 24). *What Is the Consumer Price Index (CPI)?* Investopedia. https://www.investopedia.com/terms/c/consumerpriceindex.asp#citation-40

Goodfellow, I., Bengio, Y., & Courville, A. (2016). Chapter 10: Sequence Modeling: Recurrent and Recursive Nets. Deep Learning (pp. 367 - 415). *MIT Press.*

Ha, J., Kose, M., Ohnsorge, F., & Yilmazkuday, H. (2023). *What Explains Global Inflation*(p. 2). https://documents1.worldbank.org/curated/en/099417212182324899/pdf/IDU04d2b00a8052e30403409fc1033a96b1c81ac.pdf

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735-1780. DOI: 10.1162/neco.1997.9.8.1735

Hyndman, R. J., Athanasopoulos, G. (2018). *Forecasting: Principles and Practice*. (2nd ed). Monash University, Australia. https://otexts.com/fpp2/

Keras. (n.d.). *KerasTuner*. https://keras.io/keras_tuner/

Liu, J. (2024) *Navigating the Financial Landscape: The ower and Limitations of the ARIMA Model.* Santa Monica College, https://drpress.org/ojs/index.php/HSET/article/download/19082/18645

Nyoni, T. (2019). *ARIMA modeling and forecasting of Consumer Price Index (CPI) in Germany*

(MPRA Paper No. 92442). University Library of Munich, Germany. https://mpra.ub.uni-muenchen.de/92442/

Prabhakaran, S. (2019, November 2). *Augmented Dickey Fuller Test (ADF Test)*. Machine Learning Plus. https://www.machinelearningplus.com/time-series/augmented-dickey-fuller-test/

Profillidis, V. A., & Botzoris, G. N. (2019). *Modeling of transport demand*. ScienceDirect. https://www.sciencedirect.com/science/article/pii/B9780128115138000066

Reserve Bank of Australia. (2022, August). *Statement on monetary policy: August 2022*. https://www.rba.gov.au/publications/smp/2022/aug/pdf/statement-on-monetary-policy-2022-08.pdf

Senapati, D. (2018). *Grid Search vs Random Search.* Medium. https://medium.com/@senapati.dipak97/grid-search-vs-random-search-d34c92946318

Sktime developers. (2024). *AutoARIMA: sktime 0.17.1 documentation*. Sktime. https://www.sktime.net/en/latest/api_reference/auto_generated/sktime.forecasting.arima.AutoARIMA.html

U.S. Bureau of Labor Statistics. (2024). *Consumer Price Index*. Bls.gov. https://www.bls.gov/cpi/

World Bank. (2022, January 11). *Global Growth to Slow through 2023, Adding to Risk of "Hard Landing" in Developing Economies*. World Bank. https://www.worldbank.org/en/news/press-release/2022/01/11/global-recovery-economics-debt-commodity-inequality

Winters, P. R. (1960). Forecasting sales by exponentially weighted moving averages. Management Science, 6(3), 324-342.